INTERVIEW ASSIGNMENT

Project Fund Request and Liquidation Management System

A full-stack application problem statement covering project budgets, role-based access, approval workflows, fund disbursement, expense verification, and liquidation.

Application type	Web-based business application
Primary workflow	Six stages from fund request to liquidation
User roles	Requestor, Approver, Finance, Super Admin
Core focus	Workflow logic, authorization, financial integrity, file uploads

Candidate brief • Requirements and business rules

1. Project Overview

Objective

Develop a web-based Project Fund Request and Liquidation Management System that enables employees to request funds for specific projects, obtain approvals, receive disbursements, submit expense proofs, and complete financial liquidation through a controlled six-stage workflow.

The application must support role-based access control, project and budget management, fund disbursement, expense verification, document uploads, rejection and resubmission workflows, and audit tracking.

Business Scenario

An organization manages multiple projects, each with a predefined financial allocation. Finance and Super Admin users create and manage projects and define their allocated budgets. Requestors can raise fund requests only against projects available and authorized for them.

When a requestor needs funds for a project, they submit a request specifying the amount required and select an approver. Once approved, the request moves to Finance for disbursement. After receiving the funds, the requestor spends the money on project-related activities and submits actual expenses with supporting documents such as invoices, receipts, or other proof of expenditure.

The assigned approver reviews each expense line item individually and provides final approval. Finance then verifies the approved expenses and completes liquidation. The application must maintain accurate financial records and route rejected submissions back to the correct stage.

2. User Roles and Permissions

2.1 Requestor

The Requestor is responsible for requesting project funds and submitting expense details.

- View projects available and authorized for them, and view requests in which they are involved as the requestor.
- Create a fund request against an eligible project; enter the requested amount, purpose, and supporting information.
- Select an eligible approver, view request status and approval history, and upload expense proofs with actual expenditure amounts.
- Provide a mandatory reason when submitted expenses exceed the initially requested amount.
- Correct rejected requests or expense submissions and resubmit them; view comments and rejection reasons.

Restrictions: Cannot create or modify projects, approve their own requests, disburse funds, verify expenses, or change project budget allocations. A request cannot be edited while under review unless it has been returned for correction.

2.2 Approver

- View requests assigned to them or in which they are involved; review initial fund requests and project details.
- Approve or reject initial fund requests and review expense line items individually.
- Approve or reject individual expense items, provide comments or rejection reasons, and review supporting documents.
- Provide final approval once all expense items have been reviewed; view request and expense approval history.

Restrictions: Cannot create or modify projects, disburse funds, or complete liquidation. Must not approve requests not assigned to them and must provide a rejection reason when rejecting a request or item.

The approver assigned to the initial request must also review the expense submission unless an authorized reassignment is performed before approval.

2.3 Finance

- Create and manage projects, define budgets, update project details, and activate or deactivate projects.

- View all requests across projects and users, including requested, approved, disbursed, and liquidated amounts.
- Disburse approved funds and record transaction references, dates, and notes.
- Review approved expense items and proofs, verify expenses, and complete final liquidation.
- View project balances, reassign the approver while the initial request is awaiting approval (with audit logging), and view financial history.

Restrictions: Cannot invite users unless separately granted Super Admin privileges. Cannot bypass approval stages or liquidate expenses without the required approval.

2.4 Super Admin

- Invite users, assign roles, and manage user access and account status.
- Create and manage projects, define budgets, update details, and activate or deactivate projects.
- View all requests across projects and users; reassign approvers while initial approval is pending, subject to authorization and audit logging.
- View audit logs and user activity.

Finance and Super Admin share project-management capabilities and application-wide visibility. User invitations and access administration are exclusive to Super Admin. Any financial operations by Super Admin beyond project management must be explicitly granted.

3. Project Management

3.1 Project Creation

Finance and Super Admin users are responsible for creating projects. Each project must contain:

- Project name and unique project code or identifier.
- Project description and total allocated budget.
- Project status (Active or Inactive).
- Project creation date and project creator.

3.2 Project Availability

Requestors can raise fund requests only against projects that are available and authorized for them. When creating a request, the requestor must select a project from a list of eligible projects; they cannot create a project or enter an arbitrary project name.

- Validate that the selected project exists and is active.
- Validate that the requestor is authorized to raise requests for the selected project.
- Validate the requested amount against the project's available budget and applicable reservation rules.
- Prevent new requests against inactive or unavailable projects.
- Support project-level assignment of requestors if access to specific projects must be restricted.

3.3 Project Budget Tracking

Track total allocated budget, amounts reserved for pending requests, total amount disbursed, total amount liquidated, and remaining available budget. Prevent concurrent requests from reserving more than the permitted allocation.

- Record project budget changes in the audit log.
- Preserve history and authorized access to existing requests if a project becomes inactive.
- Do not reduce allocations below existing financial commitments unless an authorized exception process is defined.

4. Six-Stage Workflow

Stage 1: Raise Fund Request — Requestor

The requestor creates a request by selecting an eligible project, entering the requested amount and purpose, adding supporting information if required, and selecting an approver. The system validates the amount against the project's available budget. If valid, the request is submitted to the approver.

- The requestor can view request details, current status, and approval history.

Status: `PENDING_APPROVAL`

Stage 2: Initial Approval — Approver

The assigned approver reviews the request and approves or rejects it. If approved, the request moves to Finance for disbursement. If rejected, it returns to Stage 1. The requestor must be able to see the rejection reason, correct the request, and resubmit it.

- A rejection reason is mandatory.

Stage 3: Fund Disbursement — Finance

Finance reviews the approved request and disburses the approved funds. Finance records the disbursed amount, date, transaction reference, and any notes.

- After disbursement is confirmed, the request moves to Stage 4 and the requestor is notified.
- If returned or rejected, the request returns to Stage 1 for correction and resubmission through the approval workflow.
- Maintain disbursement records separately from liquidation records and prevent duplicate disbursements.

Stage 4: Submit Expense Details — Requestor

After receiving funds, the requestor submits actual expenses. Each expense line item includes a description, amount spent, expense date, optional category, supporting proof such as an invoice or receipt, and comments if applicable.

- Calculate total submitted expenditure automatically.
- Compare expenditure against the initially requested amount and the disbursed amount.
- Require a justification if total expenditure exceeds the initially requested amount.
- A justification does not automatically authorize overspending; additional approval or a budget exception may be required by business rules.
- Submit the expense report to the same approver who approved the initial request.

Status: `PENDING_EXPENSE_APPROVAL`

Stage 5: Expense Review and Final Approval — Approver

The assigned approver reviews the expense report and can approve or reject each line item individually. They can inspect descriptions, amounts, proofs, and overspending justifications, and provide comments or rejection reasons.

- Supporting documents may be optional per item if permitted by business rules; mandatory proof rules must be supported where applicable.
- Final approval is allowed only after every item has been reviewed and no unresolved rejected items remain.
- If an item is rejected, the request returns to Stage 4 for correction and resubmission.
- Previously approved items remain approved unless explicitly reopened for review.
- After all items are approved and final approval is provided, the request moves to Stage 6.

Stage 6: Expense Verification and Liquidation — Finance

Finance reviews the expense report after final approval. Finance verifies approved expense items and their supporting information, marking each item verified or returning it for correction with a reason.

- Once all required items are verified, Finance completes liquidation.
- Set the request status to LIQUIDATED, record final verified expenditure, update project financial balances, record the transaction and audit history, and notify the requestor.
- If rejected or returned, send the request to Stage 4 for correction and repeat the required approval and verification stages.
- Prevent duplicate liquidation of the same request.

5. Rejection and Resubmission Rules

Stage	Rejection or failure	Destination
Stage 1 — Requestor	Validation failure	Remain at Stage 1
Stage 2 — Approver	Initial request rejected	Stage 1
Stage 3 — Finance	Disbursement returned or rejected	Stage 1
Stage 4 — Requestor	Expense validation failure	Remain at Stage 4
Stage 5 — Approver	One or more expense items rejected	Stage 4
Stage 6 — Finance	Expense verification rejected	Stage 4

Every rejection or return action must record the reason, responsible user, timestamp, previous status, and new status. Resubmission must not bypass any required approval or verification stage.

If a request is returned to Stage 1 after funds have already been disbursed, preserve the disbursement record and prevent a second disbursement for the same financial obligation. The workflow and data model must explicitly handle this scenario.

6. Expense and Liquidation Management

A single fund request can contain multiple expense line items. Each line item must maintain its description and amount, date and category if applicable, document references, approval status, Finance verification status, comments and rejection reasons, timestamps, and user identifiers.

- Calculate total submitted expenditure automatically.
- Identify overspending and require justification when expenses exceed the initially requested amount.
- Define whether expenses can exceed the amount actually disbursed and whether excess requires additional approval.
- Maintain separate records for requested funds, approved funds, disbursed funds, actual expenses, verified expenses, and liquidated expenses.
- Define how unused disbursed funds are returned and recorded, including any balance-return transaction.

7. Functional Requirements

ID	Requirement
FR-01	Authentication and authorization: implement authentication and role-based access control.
FR-02	Project management: Finance and Super Admin can create, view, modify, activate, and deactivate projects.
FR-03	Project access control: requestors can raise requests only against active, authorized projects.
FR-04	Project budget management: track allocations, reserved amounts, disbursements, liquidations, and remaining funds.
FR-05	Fund request creation: select project, enter amount and purpose, and select approver.
FR-06	Budget validation: prevent requests exceeding available budget under reservation rules.
FR-07	Initial approval: assigned approver can approve or reject requests.
FR-08	Fund disbursement: Finance records disbursement and transaction details.
FR-09	Expense submission: requestors submit multiple line items with details and supporting proofs.
FR-10	Overspending justification: require justification when expenses exceed the initial request.
FR-11	Item-level review: approvers approve or reject individual expense items.
FR-12	Final expense approval: final approval requires review of all required items.
FR-13	Finance verification: Finance verifies approved items or returns invalid submissions.
FR-14	Final liquidation: Finance liquidates after required approval and verification.
FR-15	Balance updates: update financial balances without double-counting.
FR-16	Rejection and resubmission: route requests to the correct stage.
FR-17	Approver reassignment: Finance or Super Admin may reassign the approver while initial approval is pending, with audit logging.
FR-18	User management: Super Admin invites users, assigns roles, and manages access.
FR-19	Request visibility: Requestors and Approvers see only involved requests; Finance and Super Admin see all requests.
FR-20	Audit logging: record workflow transitions, financial transactions, and project changes.
FR-21	Notifications: notify users of submission, approval, rejection, disbursement, review, and liquidation.
FR-22	Search and filtering: filter by project, requestor, approver, status, and date.
FR-23	Duplicate transaction prevention: prevent duplicate disbursement and liquidation.
FR-24	Validation and error handling: validate inputs, uploaded files, amounts, and workflow transitions.

8. Non-Functional Requirements

Security

- Enforce role-based authorization for all protected actions.
- Prevent access to unauthorized projects, requests, and documents.
- Validate permissions on the backend for every operation.
- Store uploaded files securely.

Data Integrity

- Maintain accurate financial records and validate amounts against budget rules.

- Prevent duplicate financial transactions.
- Use transactional or equivalent consistency controls for critical financial updates.

Auditability

- Record important user actions, approval decisions, project changes, and financial transactions.
- Preserve request and expense history when records are rejected or resubmitted.

Reliability

- Handle repeated submissions and concurrent requests safely.
- Prevent invalid workflow transitions.
- Ensure failed transactions do not leave project balances inconsistent.

Maintainability and Usability

- Use modular components and services with testable workflow rules and financial calculations.
- Provide clear role-specific views, pending actions, statuses, rejection reasons, and approval history.
- Provide appropriate loading, success, and error states.

9. Expected Deliverables

The candidate must develop a working application demonstrating:

- User authentication and role-based access.
- Project creation and management by Finance and Super Admin.
- Project allocation and available-budget tracking.
- Fund request creation against authorized projects.
- Initial approval and rejection.
- Finance disbursement.
- Expense submission with supporting document uploads.
- Item-level expense approval and rejection.
- Final approval by the assigned approver.
- Finance verification and liquidation.
- Correct rejection and resubmission behavior.
- Accurate project balance calculations.
- Request status tracking and audit history.
- User invitations and role management by Super Admin.

The candidate must also provide setup instructions, sample data, and a brief explanation of the application architecture and important business rules.

10. Evaluation Criteria

- Correct implementation of the six-stage workflow.
- Correct rejection and resubmission behavior.
- Role-based access control and backend authorization.
- Project creation and project-level access restrictions.

- Data modeling for projects, requests, expenses, and transactions.
- Accuracy of financial and budget calculations.
- Individual expense approval and verification.
- Secure file upload and document access.
- Prevention of duplicate transactions.
- Handling of edge cases and invalid operations.
- Code organization, readability, and maintainability.
- Quality of the user interface, validation, error handling, and testing.

11. Optional Enhancements

- Dashboard showing pending requests and completed liquidations.
- Search, sorting, and pagination.
- Email or in-app notifications.
- Downloadable expense and liquidation reports.
- Project-level assignment of requestors.
- Configurable approval rules and multi-level approval workflows.
- Budget adjustment history.
- Export to CSV or Excel.
- Unit and integration tests for workflow transitions and financial calculations.

12. Use of AI Tools and Development Environments

Candidates are permitted to use any AI tools, large language models (LLMs), coding assistants, and integrated development environments (IDEs) while completing this assignment.

Important: Do not clear or delete the chat history or conversation used with AI tools during the assignment. The conversation history should remain available for review if requested during the interview.

Candidates should be prepared to explain their implementation, design decisions, workflow logic, and any code generated or assisted by AI tools.

13. Conclusion

The Project Fund Request and Liquidation Management System must provide a secure, auditable, and reliable process for managing project funds from initial request through final liquidation. The solution should enforce the six-stage workflow, maintain accurate financial balances, restrict actions according to user roles, and support appropriate correction and resubmission when requests or expenses are rejected.